

DE Lab2
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Q1:

a) V_{T0}

find two points above the threshold and fit a line, find V at 0A

I_{DS} vs V_{GS} (plot 1): (0.55V, 12.5uA), (0.60V, 20uA)

7.5uA per 0.05V

$$0.55 - (12.5\mu A / 7.5\mu A) * 0.05V = 0.4667V$$

$$V_{T0} \approx 0.4667V$$

b) λ

in saturation, $I_D = I_{D(sat)} (1 + \lambda V_{DS})$

$$\lambda = 1/I_D dI_D/dV_{DS} \text{ (slope(A/V) per current(A) = } V^{-1})$$

I_{DS} vs V_{DS} (plot 2): (0.6V, 235uA), (0.7V, 241uA)

0.65V, $I_D = 239\mu A$. $dI_D = 6\mu A$, $dV_{DS} = 0.1V$

$$\lambda = 60 \mu A/V / 239\mu A = 0.251 V^{-1}$$

$$\lambda = 0.251 V^{-1}$$

c) γ

$$I_D \propto (V_{GS} - V_T)^2$$

$$\sqrt{I_D} \propto (V_{GS} - V_T)$$

I_{DS} vs V_{SB} (plot 3): (0V, 265uA), (1.2V, 185uA)

$$\sqrt{265} - \sqrt{185} = 16.278 - 13.601 = 2.667$$

$$2.667/16.278 = 0.1645$$

$$V_{GS} - V_{T0} = 1.2 - 0.4667 = 0.7333V$$

$$\Delta V_T = 0.1645 * 0.7333V = 0.1206V$$

$$\Delta V_T = V_T - V_{T0} = \gamma(\sqrt{V_{SB} + |\phi_F|} - \sqrt{|\phi_F|})$$

$$0.1206V = \gamma(\sqrt{1.2 + 0.88} - \sqrt{0.88})$$

$$0.1206V = \gamma(0.5041 V^{1/2})$$

$$\gamma = 0.2392 V^{1/2}$$

d) μ_n

in linear region $I_D = \mu_n C_{ox} W/L [(V_{GS} - V_T)V_{DS} - (V_{DS}^2)/2] (1 + \lambda V_{DS})$

I_{DS} vs V_{GS} (plot 1): (1.2V, 87uA)

$$87 \times 10^{-6} A = \mu_n * 1.1 \times 10^{-6} F/cm^2 * 5 * [(1.2V - 0.4667V) * 0.1V - (0.1V^2)/2] (1 + 0.251 V^{-1} * 0.1V)$$

$$\mu_n = 87 \times 10^{-6} A / [1.1 \times 10^{-6} F/cm^2 * 5 * [(1.2V - 0.4667V) * 0.1V - (0.1V^2)/2] (1 + 0.251 V^{-1} * 0.1V)]$$

$$\mu_n = 225.8176 cm^2/Vs$$

ii) This value is a bit lower than the expected 350-500 range, I'm honestly not 100% sure why, is my λ factor lower than expected?

iii) $I_D = \mu_n C_{ox} W/L [(V_{GS} - V_T)V_{DS} - (V_{DS}^2)/2] (1 + \lambda V_{DS})$

$$I_D = 350 * 1.1 \times 10^{-6} * 5 * [(0.7333V) * 1.2V - (1.2^2)/2] (1 + 0.251 * 1.2) = 401\mu A$$

$$I_D = 401\mu A$$

e) $V_{DS(sat)}$

i) expected of long-channel

$$V_{DS(sat)} = V_{GS} - V_T = 1.2V - 0.4667V = 0.7333V$$

ii) observed from graph

I_{DS} vs V_{DS} , $V_{GS}=1.2V$ (plot 2): around $V_{DS}=0.4V$

f)

$$V_{DS(sat)} \approx E_{cL}$$

$$E_{cL} \approx 0.4V$$

i) I_{DS}

$$\text{model 3: } I_{D(sat)} = W v_{sat} C_{ox} (V_{GS} - V_T)^2 / (V_{GS} - V_T + E_{cL}) (1 + \lambda V_{DS})$$

$$\text{model 3: } I_{D(sat)} = W \mu_n E_c C_{ox} (V_{GS} - V_T)^2 / (V_{GS} - V_T + E_{cL}) (1 + \lambda V_{DS})$$

$$\text{model 2: } I_{D(sat)} = W \mu_n C_{ox} (V_{GS} - V_T)^2 / (V_{GS} - V_T + E_{cL}) (1 + \lambda V_{DS})$$

differ by a factor of $2 / (1 + (V_{GS} - V_T) / E_{cL}) = 0.58$

$$I_{Dlong} = 1/2 \mu_n C_{ox} W/L (V_{GS} - V_T)^2 (1 + \lambda V_{DS}) = 436\mu A$$

$$I_{Dshort} = I_{Dlong} \cdot 0.58 = 253\mu A$$

$$I_{DS} = 253\mu A$$

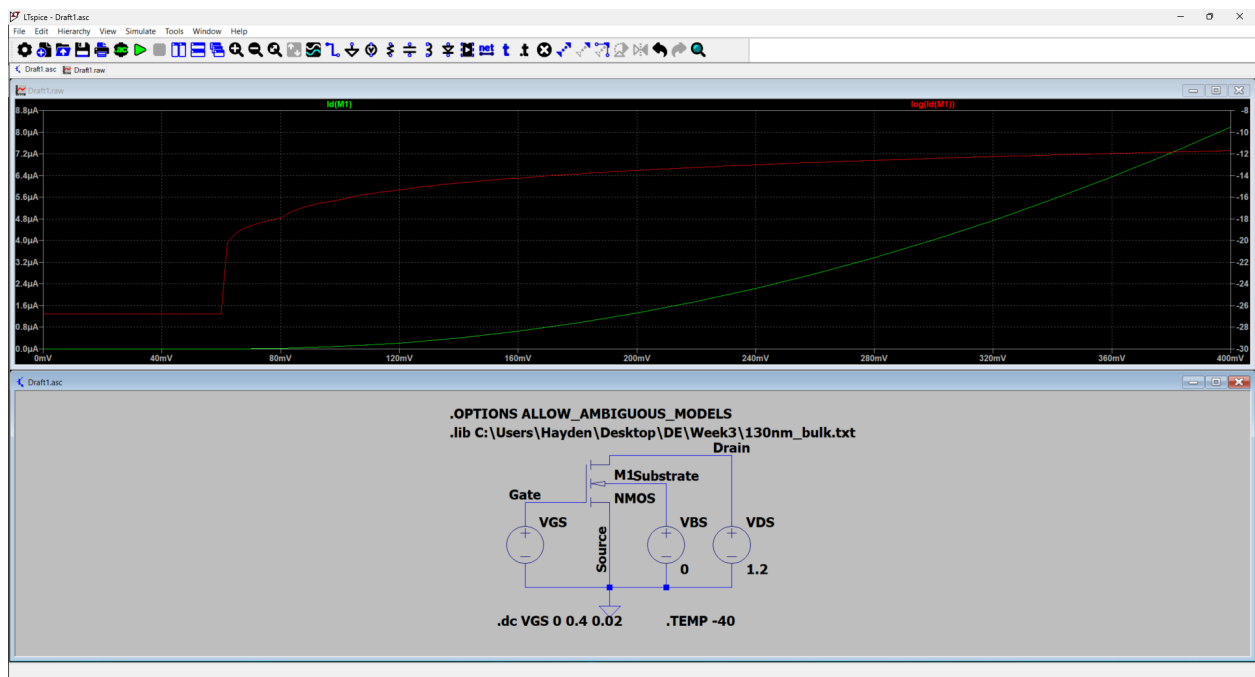
f.i.a) From either plot 2 or 3, we can see a result of $265\mu A$. Our value is 4.5% lower than this.

f.i.b) in the previous part we got $401\mu A$, this is significantly lower than that.

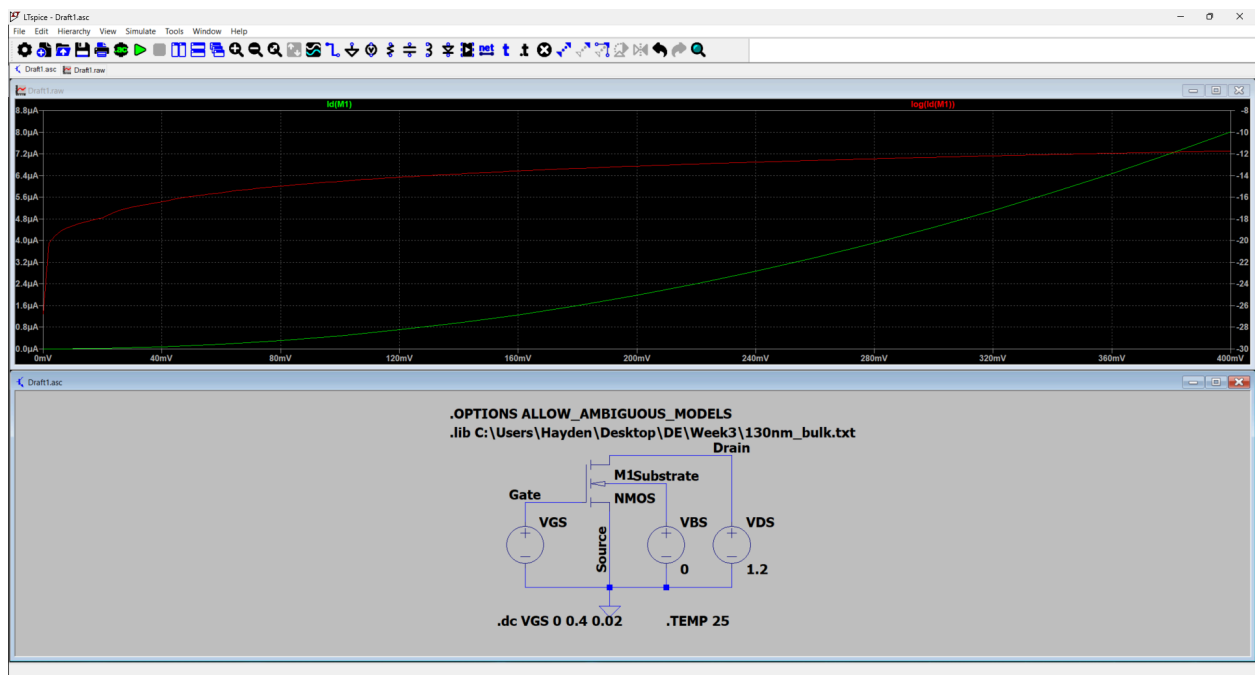
2) effect of temp on IDS vs VGS

a) IDS log IDS vs VGS 0.0-0.4

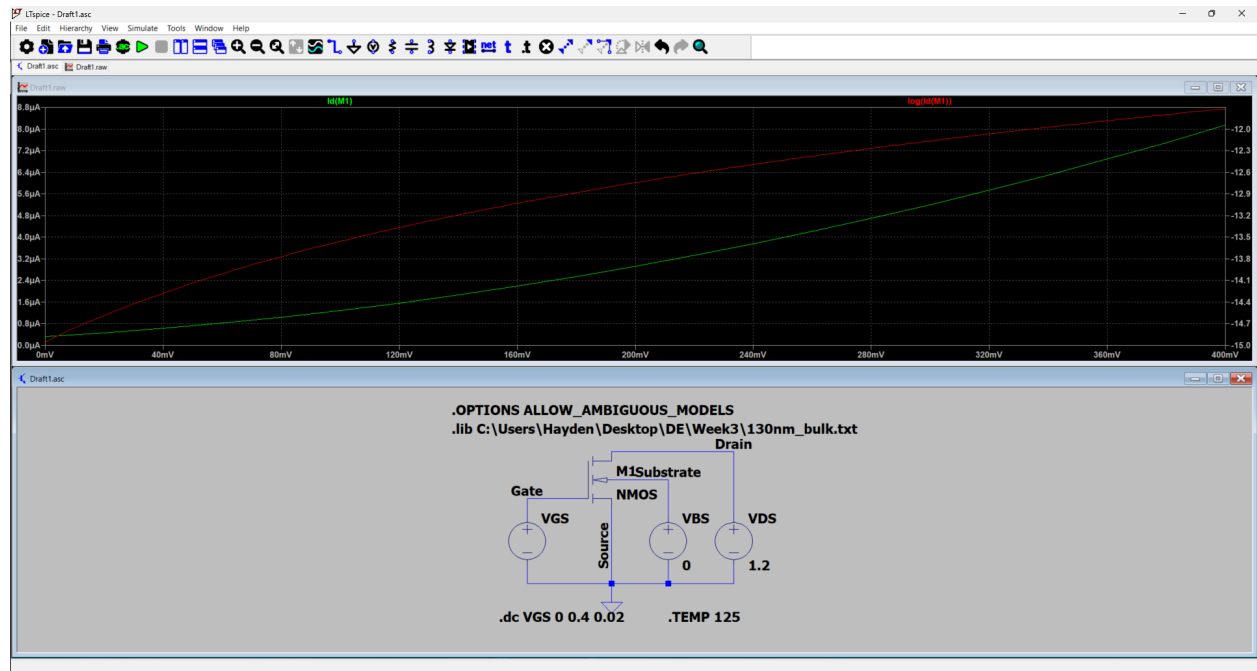
a -40C



a 25C



a 125C

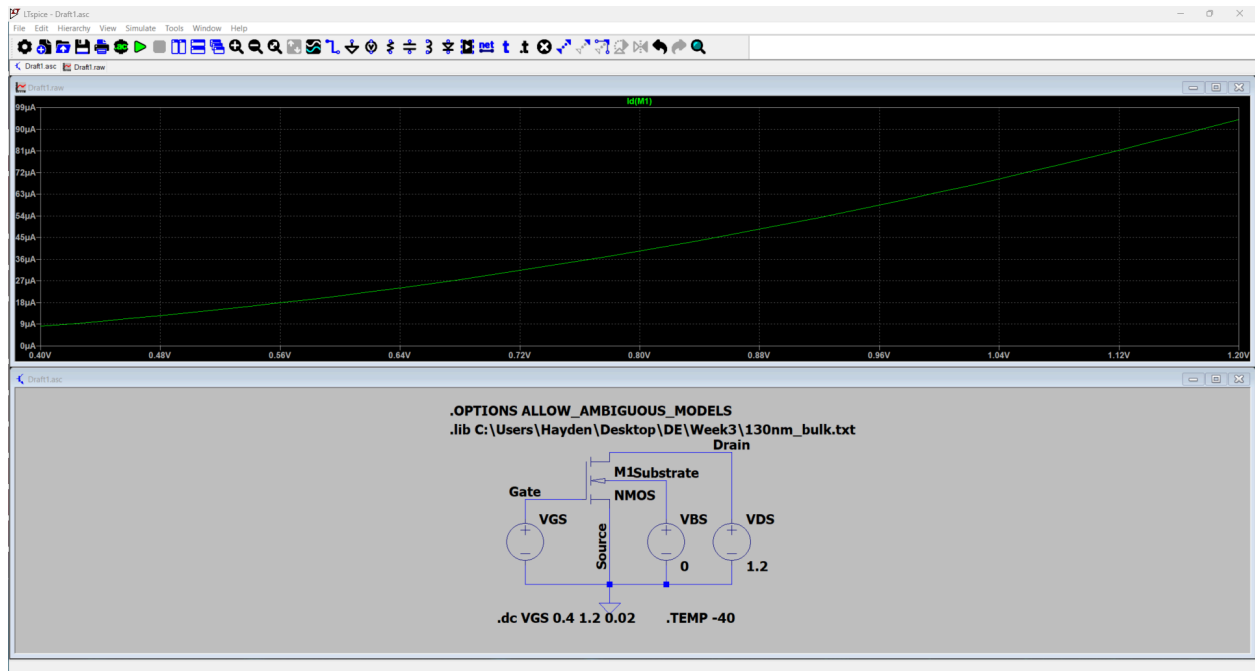


aiii)

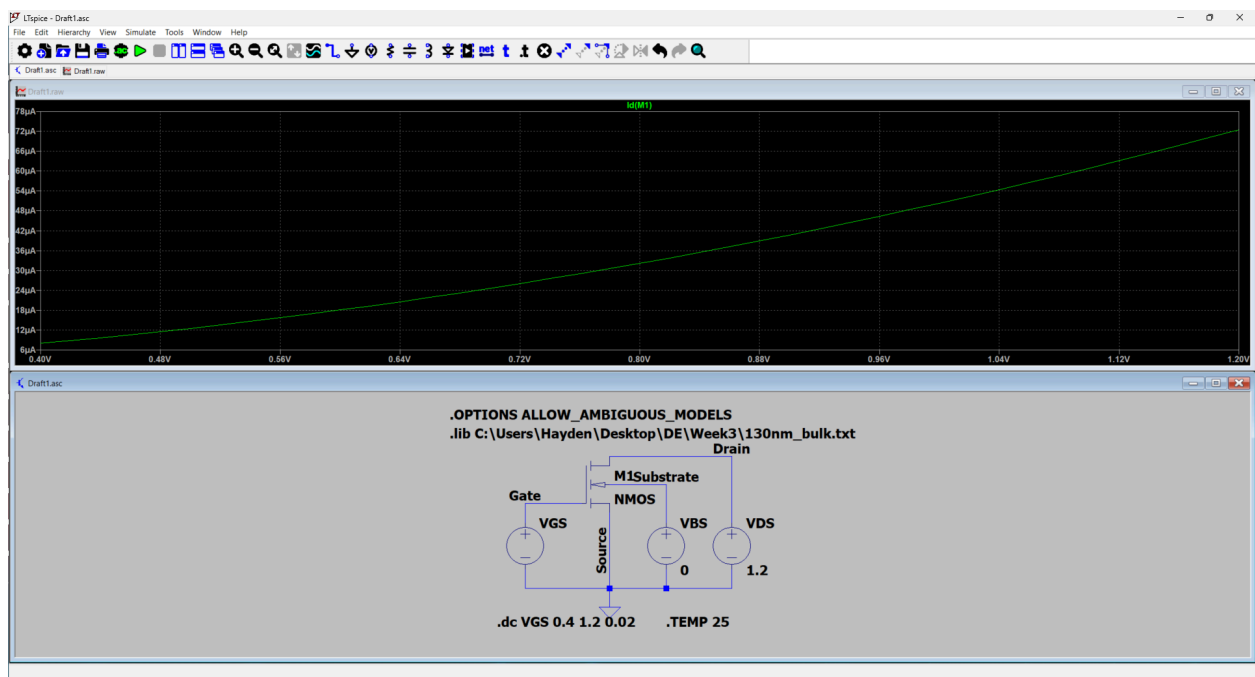
Though it's low enough to only be visible on a log scale, the current shifts upward with temperature. At -40C it starts around 10^{-27} A and only eventually jumps around 0.06V, while at 125C it already starts at 10^{-15} A. Because of this, lower temps are desirable if you want low leakage currents and low standby power consumption. This is because the subthreshold current $I_{D(sub)} \propto e^{(V_{GS} - V_{th}) / n k T}$.

b) IDS vs VGS 0.4-1.2

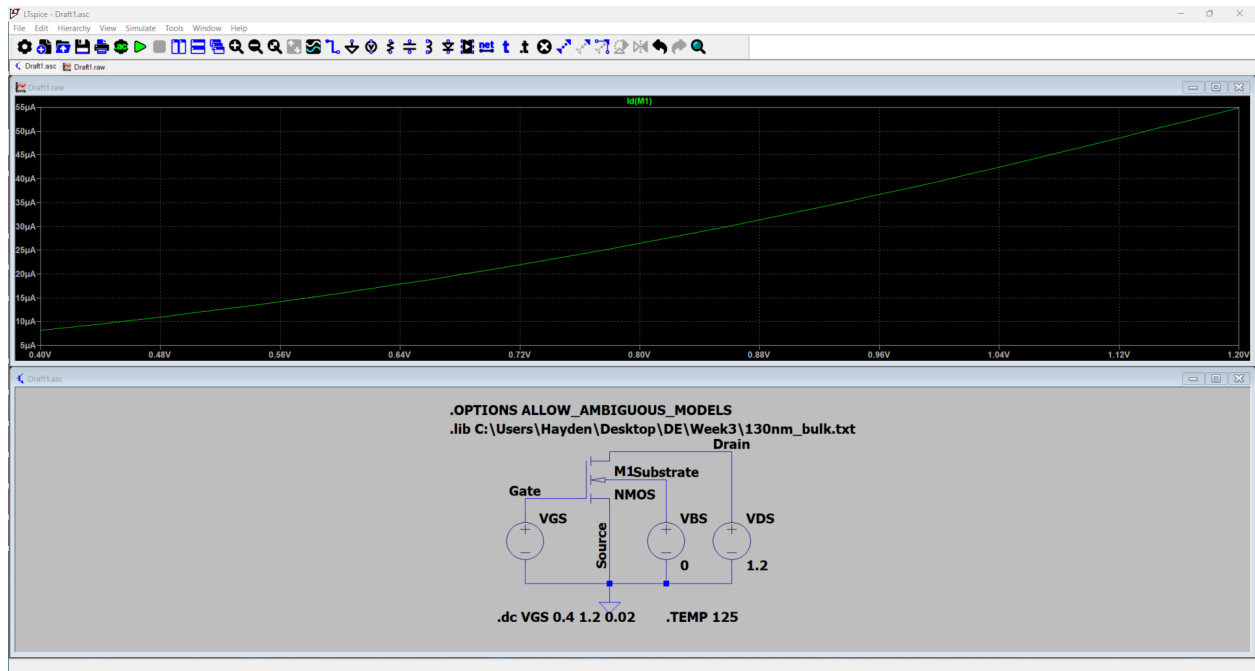
b -40C



b 25C



b 125C



biii)

The current now shifts down with temperature. At -40C current rose up to 95 μ A, while at 125C it only reached 55 μ A. Because of this, lower temps are again desirable to maximize current and performance. This is because carrier mobility μ decreases with temperature, decreasing $I_{D(sat)}$.